

Elastic wave propagation in a 3-D unbounded random heterogeneous medium coupled with a bounded medium. Application to seismic soil–structure interaction (SSSI)

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SUMMARY

We present a sub-structuring method for the coupling between a large elastic structure, and a stratified soil half-space exhibiting random heterogeneities over a bounded domain and impinged by incident waves. Both media are also weakly dissipative. The concept of interfaces classically used in sub-structuring methods is extended to ‘volume interfaces’ in the proposed approach. The random dimension of the stochastic fields modelling the heterogeneities in the soil is reduced by introducing a Karhunen–Loève expansion of these stochastic fields. The coupled overall problem is solved by Monte-Carlo simulation techniques. A realistic example of a large industrial structure interacting with an uncertain stratified soil medium under earthquake is finally presented. This case study and others validate the presented methodology and its ability to handle complex mechanical systems. Copyright © 2002 John Wiley & Sons, Ltd.

KEY WORDS: random media; wave propagation; dynamic sub-structuring; boundary element method; Karhunen–Loève expansion; Monte-Carlo simulation; soil–structure interaction

1. INTRODUCTION

Propagation phenomena arising from waves in randomly heterogeneous media can be observed in various applications, such as, non-exhaustively, light propagation in astronomical problems, polarization of electromagnetic waves in heterogeneous media, seismic waves in geophysics and seismology, or diffusion of sounds in complex geometries [1]. They have been addressed in the literature both on the analytical and, more recently, numerical aspects. From a mathematical point of view, the problem at hand belongs to the theory of boundary value problems for (linear) partial differential equations with random coefficients. In this case, the

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Contract/grant sponsor: Electricité de France, Direction Recherche et Développement

wave equation for acoustics or elastodynamics, or its counterpart in the frequency domain, is dealt with, to follow the most recent developments, by asymptotic analysis techniques. In the low-frequency range, and for not too small fluctuations of the heterogeneities in the random medium, localization phenomena dominate. This was shown, for instance, by Asch *et al.* [2] for scalar waves in stratified random media, or Pierre [3] for some simple slender structures. In the high-frequency range, and for weak fluctuations in the random medium, the wave propagation features are described very effectively by radiative transport equations for the angularly resolved energy densities (see References [4, 5]). Diffusion limits can be obtained, leading to equi-partition rules for the energies associated with each different propagation modes [4], similarly to the assumption introduced in the SEA theory for the eigenmodes of mechanical subsystems [6]. However, a general theory for transition from localization to transport and diffusion lacks at present; practically, it is impossible to decide between these two limits for applications to more complex geometries and boundary conditions. At this stage, possible alternatives are the homogenization, equivalent medium or Kramers–Krönig equation-like techniques as commonly implemented for composite materials (see for instance References [7] or [8]), but they are also restricted to very simple configurations and load shapes. Numerical simulations are used whenever the previous approaches cannot be applied. This is for instance the case in seismology by finite difference methods [9] and in structural dynamics by spectral reduction techniques and stochastic finite elements [10] or non-parametric models of symmetric positive-definite random matrices [11].

In this paper we focus on numerical methods as well in view of application to complex geometries, boundary conditions or loads, typically for wavelength larger or of the order of the system characteristic dimensions, and for large fluctuations of the mechanical parameters in the random medium. More specifically, we consider the influence of random mechanical properties of the soil on soil–structure interaction under earthquake. From an engineering point of view, this problem may be stated as follows: at a large scale (the whole site), the soil exhibits a rather deterministic horizontal stratification, whereas at the scale of the foundation it exhibits large random uncertainties. Since *in situ* measurements of its mechanical properties are usually performed in an area close to the structure to be built, a quantification of this randomness—through means and variances for instance—is available only in a bounded volume close to the foundation. Furthermore, uncertainties are reasonably expected to perturbate the foundation movement primarily when they are close to it, at first glance through possible localization. However stratification plays a significant role in the reflexion and radiation of waves, both near and relatively far away from the foundation of a large structure. Thus the modelization of the stratification at large distances, and the uncertainties near the foundation, are deemed necessary. Since analytical methods cannot be used for such a complex coupled system, numerical analysis are performed.

Having these considerations in mind, the strategy retained for the modelization is the following. Concerning the soil stratification, it is inconceivable to use finite element or finite difference methods. Indeed they would need the introduction of absorbing conditions at the boundary of the mesh formulated to radiate both the body waves and the surface waves (of the Rayleigh, Love, or even Stoneley types) typically arising in stratified media. The effects of corners and edges for such artificial boundaries should also be taken into account. Attempts have been made to write such conditions for elastic waves propagating in highly heterogeneous media [12], but an efficient and robust numerical implementation remains problematical. However numerical methods to calculate the Green function of a stratified half-space are now

well established [13] and their solutions account naturally for body and surface waves. This suggests the use of the boundary element method to modelize the stratified soil [14]. As for its bounded random part, it is modeled by the spectral stochastic finite-element method [10] in conjunction with Monte-Carlo simulations. At last structural displacements are typically obtained by, for instance, component mode synthesis, but a discussion on this issue is out of the scope of this paper. Then the coupling between the stratified soil and the structure is written following a classical sub-structuring technique which allows the use of different discretization schemes in both media. By this method the deterministic dynamic stiffnesses of the soil and the structure are condensed on their interface (which is the structural foundation) and some constraint modes need to be calculated. For the coupling between the stratified soil and its random part, it is not desirable to use such a technique. Indeed it would require the calculation, for each trial in the Monte-Carlo simulations, of the condensed stiffness of the random part on its boundary, and constraint modes which are non-linear with respect to the random soil parameters. That is why we introduce an alternative sub-structuring method for the latter coupling, essentially in order to circumvent this difficulty. Its principle has strong connection with the classical Lippmann–Schwinger equation in quantum mechanics, and the more general multi-scale coupling methods recently proposed by Hughes [15] and Ben Dhia [16]. A procedure akin to the one presented here has been applied recently by Muijres *et al.* [17] for the case of acoustic wave propagation in two-dimensional (2-D) media with randomly distributed small heterogeneities of constant shapes. The proposed methodology is oriented toward full three-dimensional (3-D) simulations in unbounded media, as is usually necessary for seismic soil–structure interaction analysis in order to account, at first, for various incidence of the seismic waves, and in the present case, for spatial variability of the soil parameters near the foundation of large structures.

This paper is organized as follows. In Section 2 we present our basic notations and formulate the overall propagation problem as a general transmission problem for a given incident wave. In Section 3 we recall the classical sub-structuring method on surface interface used to compute its solution. Then a sub-structuring method on ‘volume interface’ is proposed in Section 4 in order to account for the random heterogeneity in the soil half-space and couple it with the deterministic stratified soil–structure system. In Section 5 we present the spectral reduction technique used throughout the analysis to represent the randomness of the mechanical parameters in the soil medium. It can be seen as a (spectral) finite-element discretization along the random dimension of the initial problem, implemented in parallel with the finite-element discretization of its geometry and equilibrium equations. Then a very simple numerical test of the methodology is briefly described in Section 6, before we present a more realistic case study in Section 7: the dynamic interaction between a nuclear power plant and an uncertain stratified soil medium. We finally draw some conclusions and perspectives in Section 8.

2. MODELLING OF THE COUPLED WAVE PROPAGATION PROBLEM

2.1. Notations and general mechanical hypotheses

Let Ω_s be the half-space in $\mathbb{R}^3$ occupied by the soil, Ω_b be the bounded open domain occupied by the elastic structure, and let $\Omega = \Omega_s \cup \Omega_b$ be the overall domain. We consider throughout this paper the linearized vibrations of the structure about a static equilibrium configuration,

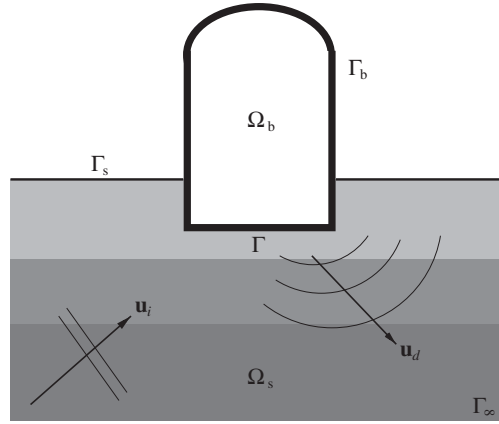


Figure 1. Model geometry of a structure occupying the bounded domain Ω_b coupled with a stratified soil occupying the half-space Ω_s and impinged by an incident wave $\mathbf{u}_i$.

when coupled with the elastic soil medium (see Figure 1). Since a linear behaviour is assumed for both media, a frequency-domain formulation can be used, the convention for the Fourier transform being $\mathbf{u}(\omega) = \int_{\mathbb{R}} e^{-i\omega t} \mathbf{u}(t) dt$ where ω denotes the circular frequency, $\mathbf{u}$ is any vector in $\mathbb{C}^3$ and $\bar{\mathbf{u}}$ its conjugate ($\mathbb{R}$ and $\mathbb{C}$ are the sets of real and complex numbers, respectively). The usual scalar product of two vectors $\mathbf{u} = \{u_1, u_2, u_3\}$ and $\mathbf{v} = \{v_1, v_2, v_3\}$ is denoted by $\mathbf{u} \cdot \mathbf{v} = \sum_j u_j v_j$ and the tensor product of two tensors $\mathbf{A} = \{a_{jk}\}$ and $\mathbf{B} = \{b_{jk}\}$, $j, k \in \{1, 2, 3\}$, is denoted by $\mathbf{A} : \mathbf{B} = \text{Tr } \mathbf{A} \mathbf{B}^T = \sum_{j,k} a_{jk} b_{jk}$ where Tr is the trace of a matrix and T stands for its transpose.

Let $\Gamma = \partial\Omega_s \cap \partial\Omega_b$ be the interaction surface between Ω_b and Ω_s , and let Γ_∞ be formally the boundary of Ω_s at infinity. We denote by $\Gamma_b = \partial\Omega_b \setminus \Gamma$ the traction-free interface between the structure and the ambient air, and by Γ_s the traction-free surface of the soil half-space. A body force field $\mathbf{f}$ is applied in the latter, which is also impinged by an incident wave denoted by $\mathbf{u}_i(\mathbf{x}, \omega)$ and defined classically in the soil half-space before it is excavated for the construction of the structural foundations. Let $\mathbf{u}(\mathbf{x}, \omega)$ be the total displacement field at each point $\mathbf{x}$ in Ω , then it satisfies the following Navier equation:

$$-\text{Div } \boldsymbol{\sigma}(\mathbf{u}) - \rho \omega^2 \mathbf{u} = \mathbf{f} \quad (1)$$

with boundary conditions $\mathbf{u} = \mathbf{u}_i$ on Γ_∞ and $\boldsymbol{\sigma}(\mathbf{u})\mathbf{n} = \mathbf{0}$ on $\Gamma_b \cup \Gamma_s$; $\rho^r(\mathbf{x})$ is the material mass density when $\mathbf{x} \in \Omega_r$, with $r = b$ or s (these superscripts are kept in the remainder of the paper to refer to Ω_b or Ω_s , respectively); $\boldsymbol{\sigma}(\mathbf{u})$ is the elastic stress tensor defined by $\boldsymbol{\sigma}^r(\mathbf{u}) = \mathbf{D}^r : \boldsymbol{\varepsilon}(\mathbf{u})$ where $\mathbf{D}^r(\mathbf{x}, \omega)$ is the fourth-order elasticity tensor of the materials when $\mathbf{x} \in \Omega_r$ and $\boldsymbol{\varepsilon}(\mathbf{u})$ is the linearized strain tensor. At last $\mathbf{n}^r$ denotes the outward unit normal vector to $\partial\Omega_r$ and the traction field on $\partial\Omega_r$ associated with the displacement field $\mathbf{u}$ is denoted by $\mathbf{t}^r(\mathbf{u}) = \boldsymbol{\sigma}^r(\mathbf{u})\mathbf{n}^r$. In Ω_b we will assume that the structure exhibits viscous damping, whereas in Ω_s we will assume that the soil medium exhibits hysteretic damping (independently from the frequency).

2.2. Transmission problem for the diffracted field

The total displacement $\mathbf{u}$ in Ω is decomposed into the incident wave $\mathbf{u}_i$ in the soil half-space, and a diffracted field $\mathbf{u}_d$ in Ω such that

$$\mathbf{u} = \mathbf{u}_i + \mathbf{u}_d \quad (2)$$

The incident wave $\mathbf{u}_i$ satisfies the homogeneous Navier equation in Ω_s together with the traction-free boundary condition on Γ_s . Thus $\mathbf{u}_d$ satisfies the Navier equation (1) with $\mathbf{u}_d = \mathbf{0}$ on Γ_∞ and the traction-free boundary condition on $\Gamma_s \cup \Gamma_b$. Note that thanks to the hysteretic damping in the soil, $\mathbf{u}_d$ can be assumed square integrable on Ω_s ; however $\mathbf{u}_i$ does not satisfy this property. In addition the total displacement and traction fields are continuous on Γ in order to ensure the kinematic and dynamic equilibria of the coupled soil–structure system. Thus, the diffracted field $\mathbf{u}_d$ is solution of the transmission problem: for all $\omega \in \mathbb{R}$, find $\mathbf{u}_d$ such that

$$\begin{aligned} -\text{Div } \boldsymbol{\sigma}(\mathbf{u}_d) - \rho\omega^2 \mathbf{u}_d &= \mathbf{f} && \text{in } \Omega_b \cup \Omega_s \\ [\mathbf{u}_d] = \mathbf{u}_i, [\mathbf{t}(\mathbf{u}_d)] &= -\mathbf{t}^s(\mathbf{u}_i) && \text{on } \Gamma \\ \mathbf{t}(\mathbf{u}_d) &= \mathbf{0} && \text{on } \Gamma_b \cup \Gamma_s \\ \mathbf{u}_d &= \mathbf{0} && \text{on } \Gamma_\infty \end{aligned} \quad (3)$$

In the above $[\mathbf{u}_d]$ is the displacement field jump on Γ defined by $[\mathbf{u}_d] = \mathbf{u}_d^b|_\Gamma - \mathbf{u}_d^s|_\Gamma$ where by definition $\mathbf{u}_d^r = \mathbf{u}_d|_{\Omega_r}$; $[\mathbf{t}(\mathbf{u}_d)]$ is the traction field jump on Γ oriented by the outward unit normal vector $\mathbf{n}^b$ to the structure domain Ω_b such that $[\mathbf{t}(\mathbf{u}_d)] = \mathbf{t}^b(\mathbf{u}_d) + \mathbf{t}^s(\mathbf{u}_d)$. Introducing a variational formulation of the transmission problem (3) above it can be proven [18, 19] that this boundary-value problem has a unique solution in a proper set of admissible displacement fields satisfying the linear constraint $[\mathbf{u}_d] = \mathbf{u}_i$, for the data $\mathbf{f}$ in Ω_s , and $\mathbf{u}_i$ and $\mathbf{t}^s(\mathbf{u}_i)$ on Γ in proper functional spaces. Therefore, there exist linear continuous operators $\mathcal{U}(\omega)$ and $\mathcal{Q}_i(\omega)$ such that this solution writes

$$\mathbf{u}_d = \mathcal{Q}_i \mathbf{u}_i + \mathcal{U} \mathbf{f} \quad (4)$$

Note that $\mathcal{Q}_i$ (sometimes called scattering operator in the literature) is defined for $\mathbf{u}_i$ on Γ , and that $\mathcal{U}$ is symmetric. Also note that the jump condition $\mathbf{t}^s(\mathbf{u}_i)$ does not appear explicitly in the above equation since this traction field can be uniquely determined from the knowledge of $\mathbf{u}_i$ on Γ , provided that the latter satisfies the Navier equation in Ω_s . These operators can be formally constructed by boundary integral equations using the Green function of the coupled soil–structure system in Ω , as done in Reference [19]. However we start by showing how the solution of the general transmission problem above is practically obtained by a classical sub-structuring technique for a stratified soil medium, first ignoring its bounded random part underneath the foundation.

3. DYNAMIC SUB-STRUCTURING ON SURFACE INTERFACE

In this section we present the classical dynamic sub-structuring method on surface interface used to compute the solution of the transmission problem (3). The soil medium is assumed

strictly stratified and the uncertainties on its mechanical parameters are ignored. The dynamic sub-structuring technique presented here uses the fixed-interface eigenmodes of the structure [20, 21] and a reduced matrix model for its dynamic stiffness. It consists in two steps: first solve local problems in each sub-domain, namely the structure and the soil, then solve a global problem corresponding to the dynamic equilibrium of the interface between the sub-domains. One of the major advantage of this method lies in its ability to couple different discretization schemes for different sub-domains. For instance, in the present case, the soil is modeled by the boundary element method, while the structure is analysed by component mode synthesis in conjunction with a finite-element discretization.

3.1. Local problems in each sub-domain

We introduce an arbitrary displacement field ϕ defined on Γ , and solve Dirichlet's boundary-value local problems in Ω_b and Ω_s , with the boundary conditions $\mathbf{u}^b|_\Gamma = \phi$ and $\mathbf{u}^s|_\Gamma = \phi$, respectively. That is, for all $\omega \in \mathbb{R}$, find $\mathbf{u}^b(\phi)$ such that

$$\begin{aligned} \mathbf{Div} \boldsymbol{\sigma}^b(\mathbf{u}^b(\phi)) + \rho^b \omega^2 \mathbf{u}^b(\phi) &= \mathbf{0} \quad \text{in } \Omega_b \\ \mathbf{u}^b(\phi) &= \phi \quad \text{on } \Gamma \\ \mathbf{t}^b(\mathbf{u}^b(\phi)) &= \mathbf{0} \quad \text{on } \Gamma_b \end{aligned}$$

and find $\mathbf{u}^s(\phi)$ such that

$$\begin{aligned} \mathbf{Div} \boldsymbol{\sigma}^s(\mathbf{u}^s(\phi)) + \rho^s \omega^2 \mathbf{u}^s(\phi) &= \mathbf{0} \quad \text{in } \Omega_s \\ \mathbf{u}^s(\phi) &= \phi \quad \text{on } \Gamma \\ \mathbf{t}^s(\mathbf{u}^s(\phi)) &= \mathbf{0} \quad \text{on } \Gamma_s \\ \mathbf{u}^s(\phi) &= \mathbf{0} \quad \text{on } \Gamma_\infty \end{aligned}$$

The contribution from the incident wave $\mathbf{u}_i$ and the body force $\mathbf{f}$ is obtained by solving a third problem in Ω_s : Find $\mathbf{u}_{d0}$ such that

$$\begin{aligned} -\mathbf{Div} \boldsymbol{\sigma}^s(\mathbf{u}_{d0}) - \rho^s \omega^2 \mathbf{u}_{d0} &= \mathbf{f} \quad \text{in } \Omega_s \\ \mathbf{u}_{d0} &= -\mathbf{u}_i \quad \text{on } \Gamma \\ \mathbf{t}^s(\mathbf{u}_{d0}) &= \mathbf{0} \quad \text{on } \Gamma_s \\ \mathbf{u}_{d0} &= \mathbf{0} \quad \text{on } \Gamma_\infty \end{aligned}$$

3.1.1. Structure sub-domain. The solution $\mathbf{u}^b(\phi)$ of the first local problem in Ω_b is written as the sum of a dynamic part decomposed on the basis of the fixed-interface eigenmodes of the structure, and a static part $\mathbf{u}_{\text{stat}}^b = \mathcal{S}^b \phi$ where $\mathcal{S}^b$ is the elastostatic lifting operator from Γ to Ω_b . The first M_A fixed-interface eigenmodes of the structure are denoted by $\{\varphi_A\}_{1 \leq A \leq M_A}$ with the corresponding eigenvalues $\{\omega_A\}_{1 \leq A \leq M_A}$ and critical viscous damping rates $\{\zeta_A\}_{1 \leq A \leq M_A}$; they are normalized, for instance, with respect to the mass. Then we write

$$\mathbf{u}^b(\phi)(\mathbf{x}, \omega) = \mathcal{S}^b \phi(\mathbf{x}, \omega) + \sum_{A=1}^{M_A} \alpha_A(\omega) \varphi_A(\mathbf{x}) \quad (5)$$

3.1.2. Soil sub-domain. The radiated field $\mathbf{u}^s(\phi)$ solution of the second local problem in Ω_s is obtained by boundary integral equations [22]. Let $\mathbf{u}^{Gs}$ be the Green function of the half-space occupied by the stratified soil, then by definition $\mathbf{u}^{Gs}(\mathbf{x}, \mathbf{y}; \mathbf{a})$ is the displacement observed at $\mathbf{y}$ in Ω_s when a unit point force is applied at $\mathbf{x}$ along the direction of $\mathbf{a}$. It is constructed in the present case by the reflexion/transmission coefficients method due to Kennett [13]. Let $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ be an orthonormal basis of $\mathbb{R}^3$, then the first Green tensor $\mathbf{U}^{Gs}$ of the stratified soil medium is defined by

$$[\mathbf{U}^{Gs}(\mathbf{x}, \mathbf{y})]_{jk} = \mathbf{u}^{Gs}(\mathbf{x}, \mathbf{y}; \mathbf{e}_j) \cdot \mathbf{e}_k, \quad j, k \in \{1, 2, 3\}$$

We also denote by $\mathbf{T}^{Gs}$ the second Green tensor such that for any $\mathbf{x} \in \Gamma$ and $\mathbf{y} \in \Omega_s$:

$$[\mathbf{T}^{Gs}(\mathbf{x}, \mathbf{y})]_{jk} = \boldsymbol{\sigma}^s(\mathbf{u}^{Gs}(\mathbf{x}, \mathbf{y}; \mathbf{e}_j)) : (\mathbf{e}_k \otimes \mathbf{n}^s(\mathbf{y}))$$

It is computed together with the Green function by the same method, and satisfies the traction-free boundary condition on the surface Γ_s of the half-space. Then $\mathbf{u}^s(\phi)$ is given by the representation integral equation

$$\mathbf{u}^s(\phi)(\mathbf{x}, \omega) = \int_{\Gamma} \mathbf{U}^{Gs}(\mathbf{x}, \mathbf{y}) \mathbf{t}^s(\phi)(\mathbf{y}) dS(\mathbf{y}) - \int_{\Gamma} \mathbf{T}^{Gs}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{y}) dS(\mathbf{y}), \quad \mathbf{x} \in \Omega_s \quad (6)$$

where the traction field $\mathbf{t}^s(\phi)$ on Γ is solution of the boundary integral equation

$$\int_{\Gamma} \mathbf{U}^{Gs}(\mathbf{x}, \mathbf{y}) \mathbf{t}^s(\phi)(\mathbf{y}) dS(\mathbf{y}) = \kappa \phi(\mathbf{x}) + \int_{\Gamma} \mathbf{T}^{Gs}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{y}) dS(\mathbf{y}), \quad \mathbf{x} \in \Gamma \quad (7)$$

The diffracted field $\mathbf{u}_{d0}$ solution of the third local problem in Ω_s is given by the representation integral equation

$$\begin{aligned} \mathbf{u}_{d0}(\mathbf{x}, \omega) = & \int_{\Gamma} \mathbf{U}^{Gs}(\mathbf{x}, \mathbf{y}) \mathbf{t}^s(\mathbf{u}_{d0})(\mathbf{y}) dS(\mathbf{y}) + \int_{\Gamma} \mathbf{T}^{Gs}(\mathbf{x}, \mathbf{y}) \mathbf{u}_i(\mathbf{y}) dS(\mathbf{y}) \\ & + \int_{\Omega_s} \mathbf{U}^{Gs}(\mathbf{x}, \mathbf{y}) \mathbf{f}(\mathbf{y}) d\mathbf{y}, \quad \mathbf{x} \in \Omega_s \end{aligned} \quad (8)$$

where the traction field $\mathbf{t}^s(\mathbf{u}_{d0})$ on Γ is solution of the boundary integral equation:

$$\begin{aligned} \int_{\Gamma} \mathbf{U}^{Gs}(\mathbf{x}, \mathbf{y}) \mathbf{t}^s(\mathbf{u}_{d0})(\mathbf{y}) dS(\mathbf{y}) = & -\kappa \mathbf{u}_i(\mathbf{x}) - \int_{\Gamma} \mathbf{T}^{Gs}(\mathbf{x}, \mathbf{y}) \mathbf{u}_i(\mathbf{y}) dS(\mathbf{y}) \\ & + \int_{\Omega_s} \mathbf{U}^{Gs}(\mathbf{x}, \mathbf{y}) \mathbf{f}(\mathbf{y}) d\mathbf{y}, \quad \mathbf{x} \in \Gamma \end{aligned} \quad (9)$$

The free term κ is $\kappa = \frac{1}{2}$ in each layer for a grounded foundation, and $\kappa = 1$ for a superficial foundation. Regularized versions of the boundary integral equations above have been considered in Reference [14] to remove the strong singularity in the integrals whose kernels are $\mathbf{T}^{Gs}$; they also circumvent the calculation of the free term at the interface between two layers needed if Γ crosses it. Equations (7) and (9) are solved using a standard boundary element method with piecewise constant shape functions for the unknown traction fields $\mathbf{t}^s(\phi)$ and $\mathbf{t}^s(\mathbf{u}_{d0})$, and a collocation technique at the centre of each element [22].

3.2. Global problem

The displacement field ϕ is obtained writing the dynamic equilibrium of the interface Γ (global problem)

$$\mathbf{t}^b(\mathbf{u}^b(\phi)) + \mathbf{t}^s(\mathbf{u}^s(\phi)) + \mathbf{t}^s(\mathbf{u}_i) + \mathbf{t}^s(\mathbf{u}_{d0}) = \mathbf{0} \quad (10)$$

in a weak form. Expanding ϕ on a given basis $\{\phi_m\}_{1 \leq m \leq M}$ of the admissible displacement fields on Γ

$$\phi(\mathbf{x}, \omega) = \sum_{m=1}^M q_m(\omega) \phi_m(\mathbf{x})$$

the generalized co-ordinates $\boldsymbol{\alpha} = \{\alpha_1, \dots, \alpha_M\}^T$ and $\mathbf{q} = \{q_1, \dots, q_M\}^T$ are shown to be solution of the algebraic system

$$\begin{bmatrix} \mathcal{Z}^b(\omega) & \mathcal{A}(\omega) \\ \mathcal{A}(\omega)^T & Z_\Gamma^b(\omega) + Z_\Gamma^s(\omega) \end{bmatrix} \begin{pmatrix} \boldsymbol{\alpha} \\ \mathbf{q} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{F} \end{pmatrix} \quad (11)$$

where $Z_\Gamma^b(\omega)$ is the Guyan static condensation on Γ of the mass, stiffness [23] and damping matrices of the structure constructed using the elastostatic lifting operator $\mathcal{S}^b$; $Z_\Gamma^s(\omega)$ is the impedance matrix of the soil seen from Γ whose elements are

$$[Z_\Gamma^s(\omega)]_{mn} = \int_{\Gamma} \mathbf{t}^s(\phi_n) \cdot \phi_m \, dS$$

$\mathcal{Z}^b(\omega)$ is the dynamic stiffness matrix of the structure projected on its eigenmodes, which is diagonal assuming that they diagonalize the damping matrix as well

$$\mathcal{Z}^b(\omega) = \text{diag}(-\omega^2 + 2i\zeta_A \omega_A \omega + \omega_A^2)$$

$\mathcal{A}(\omega)$ is the matrix of the classical participation factors whose elements are

$$[\mathcal{A}(\omega)]_{Am} = -\omega^2 \int_{\Omega_b} \rho^b \phi_A \cdot \mathcal{S}^b \phi_m \, d\mathbf{x}$$

where the elastostatic fields $\mathcal{S}^b \phi_m$ have been called static modes, or constraint modes, in the literature. Lastly, the forcing vector $\mathbf{F}(\omega)$ for seismic waves and body forces has elements

$$F_m(\omega) = - \int_{\Gamma} \mathbf{t}^s(\mathbf{u}_i + \mathbf{u}_{d0}) \cdot \phi_m \, dS$$

Once Equation (11) above has been solved, the diffracted displacement field in the soil, for instance, is reconstructed by

$$\mathbf{u}_d^s = \mathbf{u}_{d0} + \sum_{m=1}^M q_m \mathbf{u}^s(\phi_m)$$

where $\mathbf{u}_{d0}$ and $\mathbf{q}$ (as well as $\boldsymbol{\alpha}$) depend linearly on $\mathbf{u}_i$ and $\mathbf{f}$ through Equations (8) and (11), respectively. This exhibits a formal construction of operators $\mathcal{Q}_i$ and $\mathcal{U}$.

Dual formulations have also been derived by such a kind of analysis. In these cases, an arbitrary traction field defined on Γ is introduced, Neumann's boundary-value local problems

are solved in Ω_b and Ω_s , and the global problem corresponds to the kinematic equilibrium of the interface. This is done for instance in McNeal's component mode synthesis method [24] using attachment modes, or the FETI method with Lagrange multipliers [25] for conservative structures.

3.3. Consideration of bounded soil heterogeneities

The sub-structuring method written above for two sub-domains can be extended to multi-domain problems without any difficulty (see for instance Reference [26] and references therein). In particular, it could be used to account for a bounded soil heterogeneity embedded in the surrounding stratified half-space underneath the structural foundation [26]. However this technique, as shown above for the case of a structural sub-domain, would require the calculation for the heterogeneity sub-domain of, firstly, static condensations of its stiffness and mass matrices, and secondly, some static, or constraint, modes. The latter fields are highly non-linear with respect to the mechanical characteristics of the heterogeneity, and no simple analytical relations can be found in the general case. Thus for random parameters, it would be necessary to perform an independent calculation for each of their realizations among a given population, leading to prohibitive computational costs by the present approach. This shows that, practically, the classical sub-structuring method can be applied only for deterministic heterogeneities; adapted techniques need to be implemented for the random ones, especially their coupling with unbounded media.

A second solution would consist in keeping a full finite-element model of the heterogeneity without condensation, including the background stratification and its fluctuations. Thus, the corresponding mesh of the bounded area it occupies would have to cope with both the deterministic, mean soil medium (the stratification) and its random perturbations, which can have very different length scales. The proposed methodology circumvents the difficulties inherent to the classical sub-structuring method as outlined above, and considers only the finite-element discretization of perturbations around the mean parameters of the soil medium. It is presented in detail in the next section.

4. SUB-STRUCTURING METHOD ON A VOLUME INTERFACE

By the above classical dynamic sub-structuring technique, we are now able to construct the solution of the transmission problem (3) for a deterministic stratified soil medium using the numerical Green function of a stratified half-space. Therefore, we can formally construct the operators $\mathcal{U}$ and $\mathcal{Q}_i$ introduced above. In this section, we show how to couple the deterministic stratified soil-structure system with an heterogeneity of the soil occupying a bounded domain, in such a way that systematic computations can be performed. Before presenting the sub-structuring technique used in this respect, we first start by introducing how the soil random heterogeneities are defined and accounted for.

4.1. Definition of the random heterogeneous medium in the soil

The elasticity tensor $\mathbf{D}^s$ and density ρ^s of the soil are assumed deterministic everywhere in Ω_s , except over a bounded sub-domain of Ω_s , hereafter denoted by Ω_δ (see Figure 2). They

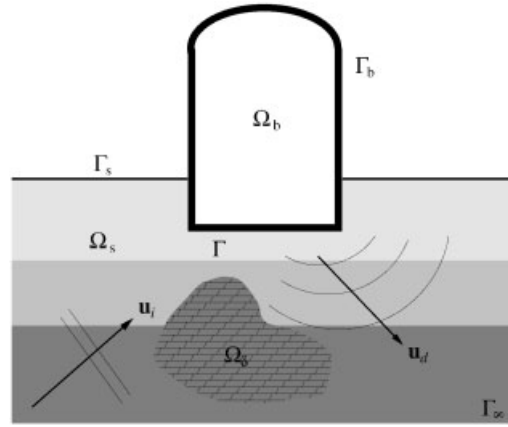


Figure 2. A structure coupled with a stratified soil half-space including random perturbations of its mechanical parameters over a bounded domain Ω_δ .

are written for all $\mathbf{x} \in \Omega_s$:

$$\mathbf{D}^s(\mathbf{x}) = \underline{\mathbf{D}}^s(\mathbf{x}) + \delta \mathbf{D}^s(\mathbf{x}) \quad (12)$$

$$\rho^s(\mathbf{x}) = \underline{\rho}^s(\mathbf{x}) + \delta \rho^s(\mathbf{x}) \quad (13)$$

where $\underline{\mathbf{D}}^s$ and $\underline{\rho}^s$ are the deterministic parts of the elasticity tensor and density of the soil, respectively, corresponding to its known stratification; and $\delta \mathbf{D}^s$ and $\delta \rho^s$ are their random perturbations—not necessarily small—over Ω_s , which are equal to zero almost surely (a.s.) in $\Omega_s \setminus \Omega_\delta$. In conjunction with these definitions, we note $\boldsymbol{\sigma}^s(\mathbf{u}) = \underline{\mathbf{D}}^s : \boldsymbol{\varepsilon}(\mathbf{u})$ and $\delta \boldsymbol{\sigma}^s(\mathbf{u}) = \delta \mathbf{D}^s : \boldsymbol{\varepsilon}(\mathbf{u})$.

The perturbations are such that the usual positiveness and symmetry properties are still satisfied for $\mathbf{D}^s$ and ρ^s : for instance, we will have $\delta \rho^s(\mathbf{x}) > -\underline{\rho}^s(\mathbf{x})$ a.s. for all $\mathbf{x} \in \Omega_\delta$. They are modelized by non-homogeneous zero-mean, second-order stochastic fields indexed on Ω_δ and equal to zero a.s. on the boundary $\partial \Omega_\delta$ of Ω_δ . We rearrange all non-zero, independent coefficients in the upper triangular part of $\delta \mathbf{D}^s$ in a random vector $\mathbf{P}^s(\mathbf{x})$ and we assume that the auto- and cross-correlation matrices of $\mathbf{P}^s$ and $\delta \rho^s$ are known. They are denoted by $\mathbf{R}_P$, R_ρ and $\mathbf{R}_{\rho, P}$ which are functions defined on $\Omega_\delta \times \Omega_\delta$ and such that

$$\mathbf{R}_P(\mathbf{x}, \mathbf{x}') = \mathbb{E}[\mathbf{P}^s(\mathbf{x}) \otimes \bar{\mathbf{P}}^s(\mathbf{x}')]]$$

$$R_\rho(\mathbf{x}, \mathbf{x}') = \mathbb{E}[\delta \rho^s(\mathbf{x}) \delta \rho^s(\mathbf{x}')]]$$

$$\mathbf{R}_{\rho, P}(\mathbf{x}, \mathbf{x}') = \mathbb{E}[\mathbf{P}^s(\mathbf{x}) \delta \rho^s(\mathbf{x}')]]$$

where $\mathbb{E}$ stands for mathematical expectation. The random dynamic stiffness associated to these perturbation fields is denoted by $\delta \mathcal{Z}(\omega)$ and will be constructed by stochastic finite elements (see Section 5). This operator is not invertible since the perturbations introduced do not satisfy *a priori* the positiveness properties of $\mathbf{D}^s$ and ρ^s (or $\underline{\mathbf{D}}^s$ and $\underline{\rho}^s$).

At last we state that, for the remaining of the paper, an underlined quantity will refer to the stratified soil mechanical parameters, excluding their random perturbations. For instance, the operators $\underline{\mathcal{U}}$ (symmetric as well) and $\underline{\mathcal{Q}}_i$ are those obtained by solving the classical sub-structuring problem (11) for a stratified soil with its mechanical parameters being $\underline{\mathbf{D}}^s$ and $\underline{\rho}^s$ everywhere in Ω_s .

4.2. Continuum-based variational formulation of the coupling

For the sake of simplicity we assume here that $\partial\Omega_\delta \cap \Gamma = \emptyset$ and $\mathbf{f} = \mathbf{0}$. The coupling between Ω and the bounded domain Ω_δ occupied by the random soil heterogeneity with the perturbation fields $\delta\mathbf{D}^s$ and $\delta\rho^s$ is written following the same general procedure as in the classical coupling case on surface interface, introducing an arbitrary force field defined all over Ω_δ . This field corresponds to the additional inertia and strain forces induced by the perturbations of the stratified soil mechanical parameters: if $\mathbf{u}_\delta$ is the total displacement field in Ω_δ when the stratified soil exhibits a random heterogeneous part over Ω_δ , then $\Phi \equiv \text{Div } \delta\sigma^s(\mathbf{u}_\delta) + \delta\rho^s\omega^2\mathbf{u}_\delta$ and the diffracted field $\mathbf{u}_d$ now satisfies the Navier equation

$$-\text{Div } \underline{\sigma}^s(\mathbf{u}_d) - \underline{\rho}^s\omega^2\mathbf{u}_d = \Phi \quad \text{in } \Omega_s$$

Φ will play in the proposed methodology the same role as ϕ in the classical sub-structuring method; however since it is defined on a volume, Ω_δ in the present case, we will refer to a sub-structuring method on a ‘volume’ interface. The two sub-domains considered are Ω with deterministic parameters in the soil, and Ω_δ with random parameters.

4.2.1. Local problems in each sub-domain. Let us consider the following two local problems in Ω and Ω_δ , respectively: for all $\omega \in \mathbb{R}$, find $\mathbf{u}_d(\Phi)$ such that

$$\begin{aligned} \text{Div } \sigma^b(\mathbf{u}_d) + \rho^b\omega^2\mathbf{u}_d &= \mathbf{0} && \text{in } \Omega_b \\ \text{Div } \underline{\sigma}^s(\mathbf{u}_d) + \underline{\rho}^s\omega^2\mathbf{u}_d &= -\Phi && \text{in } \Omega_s \\ [\mathbf{u}_d] = \mathbf{u}_i, \quad [\mathbf{t}(\mathbf{u}_d)] &= -\underline{\mathbf{t}}^s(\mathbf{u}_i) && \text{on } \Gamma \\ \mathbf{t}^b(\mathbf{u}_d) &= \mathbf{0} && \text{on } \Gamma_b \\ \underline{\mathbf{t}}^s(\mathbf{u}_d) &= \mathbf{0} && \text{on } \Gamma_s \\ \mathbf{u}_d &= \mathbf{0} && \text{on } \Gamma_\infty \end{aligned} \tag{14}$$

and find $\mathbf{u}_\delta(\Phi)$ such that

$$\begin{aligned} \text{Div } \delta\sigma^s(\mathbf{u}_\delta) + \delta\rho^s\omega^2\mathbf{u}_\delta &= \Phi && \text{in } \Omega_\delta \\ \delta\sigma^s(\mathbf{u}_\delta)\mathbf{n} &= \mathbf{0} && \text{on } \partial\Omega_\delta \end{aligned} \tag{15}$$

where $\mathbf{n}$ is the outward unit normal to $\partial\Omega_\delta$. The arbitrary force field Φ which ensures the *kinematic* equilibrium between Ω and Ω_δ is given by the kinematic compatibility condition (global problem): Find Φ such that

$$\mathbf{u}_i|_{\Omega_\delta} + \mathbf{u}_d(\Phi)|_{\Omega_\delta} - \mathbf{u}_\delta(\Phi) = \mathbf{0} \quad \text{on } \Omega_\delta \tag{16}$$

Deterministic sub-domain: From Section 2.2 we know that the first local problem (14) has a unique solution which writes

$$\mathbf{u}_d(\Phi) = \underline{\mathcal{U}}_i \mathbf{u}_i + \underline{\mathcal{U}} \Phi \quad (17)$$

Random sub-domain: Multiplying Equation (15) by any test field (weighting function) $\mathbf{v}_\delta$ defined over Ω_δ and integrating by parts leads to

$$(\delta \mathcal{L} \mathbf{u}_\delta, \mathbf{v}_\delta)_{\Omega_\delta} \stackrel{\text{def}}{=} -\omega^2 \int_{\Omega_\delta} \delta \rho^s \mathbf{u}_\delta \cdot \mathbf{v}_\delta \, d\mathbf{x} + \int_{\Omega_\delta} \delta \boldsymbol{\sigma}^s(\mathbf{u}_\delta) : \boldsymbol{\varepsilon}(\mathbf{v}_\delta) \, d\mathbf{x} = \int_{\Omega_\delta} \Phi \cdot \mathbf{v}_\delta \, d\mathbf{x} \quad (18)$$

This system is usually not invertible, since the mass and stiffness operators involved are not positive-definite as seen in Section 4.1. In order to overcome this difficulty, we rather write the solution of the second local problem (15), using (17) in (16), in the form

$$\mathbf{u}_\delta(\Phi) = \mathbf{u}_i + \underline{\mathcal{U}}_i \mathbf{u}_i + \underline{\mathcal{U}} \Phi \quad (19)$$

in the weak sense on Ω_δ . Then we use Equation (18) to write the *dynamic* equilibrium between Ω and Ω_δ and solve the global problem for Φ corresponding to a dynamic compatibility condition in lieu of the above kinematic one, Equation (16).

4.2.2. Global problem. As explained above, we now inject representation (19) into the equilibrium equation (18), in order to obtain a new equation for the unknown Φ . A proper variational formulation is derived by choosing the displacement test fields $\mathbf{v}_\delta$ in the range space of the operator $\underline{\mathcal{U}}$ as $\mathbf{v}_\delta = \underline{\mathcal{U}} \Psi$ for a body force field Ψ defined on Ω_δ . We thus obtain the global problem, using the symmetry of $\underline{\mathcal{U}}$:

$$[\underline{\mathcal{U}} - \underline{\mathcal{U}} \delta \mathcal{L} \underline{\mathcal{U}}] \Phi = \underline{\mathcal{U}} \delta \mathcal{L} [\mathcal{I} + \underline{\mathcal{U}}_i] \mathbf{u}_i \quad (20)$$

where $\mathcal{I}$ is the identity operator for Ω_s . Once the force field Φ has been computed through the algebraic system (20) above, the diffracted random field $\mathbf{u}_d$ in Ω with the actual mechanical parameters, is reconstructed using the operator $\underline{\mathcal{U}}$. The total displacement field is obtained adding the incident wave $\mathbf{u}_i$ in Ω_s :

$$\mathbf{u} = (\mathcal{I} + \underline{\mathcal{U}}_i) \mathbf{u}_i + \underline{\mathcal{U}} \Phi \quad (21)$$

4.3. Finite element approximation

We briefly outline a finite-element discretization of Equation (20) by the usual Ritz–Galerkin method. We expand the unknown force field Φ on a given basis $\{\Phi_n\}_{1 \leq n \leq N}$ as

$$\Phi(\mathbf{x}, \omega) = \sum_{n=1}^N q_{\delta,n}(\omega) \Phi_n(\mathbf{x})$$

where $\mathbf{q}_\delta = \{q_{\delta,1}, \dots, q_{\delta,N}\}^T$ are its complex-valued generalized coordinates. The random dynamic stiffness operator for the perturbations over Ω_δ is projected on a finite-element basis denoted by $\{\mathbf{w}_{Jj}; 1 \leq j \leq 3, 1 \leq J \leq N_J\}$ where N_J is the number of nodes. Then $\mathbf{q}_\delta$ is solution of the algebraic system

$$([\mathbf{U}] - [\mathbf{U}_\delta]^T [\delta \mathbf{Z}] [\mathbf{U}_\delta]) \mathbf{q}_\delta = \mathbf{F}_\delta \quad (22)$$

where we have noted

$$[\mathbf{U}_\delta]_{Jjn} = \underline{\mathcal{U}}\Phi_n(\mathbf{x}_J) \cdot \mathbf{e}_j, \quad [\mathbf{U}]_{mn} = \int_{\Omega_\delta} \underline{\mathcal{U}}\Phi_n \cdot \Phi_m \, d\mathbf{x}$$

and the dynamic stiffness matrix of the perturbations takes the usual form

$$[\delta\mathbf{Z}(\omega)] = -\omega^2[\delta\mathbf{M}] + [\delta\mathbf{K}], \quad [\delta\mathbf{Z}]_{JjKk} = (\delta\mathcal{Z}\mathbf{w}_{Kk}, \mathbf{w}_{Jj})_{\Omega_\delta}$$

In this scheme the test fields $\underline{\mathcal{U}}\Phi_n$ are simply interpolated at the nodes $\mathbf{x}_J$, $1 \leq J \leq N_J$, of the finite-element mesh of the random heterogeneity. These fields are smooth enough when constant shape functions Φ_n are used over the elements. Finally, the driving term is

$$\mathbf{F}_\delta = [\mathbf{U}_\delta]^T [\delta\mathbf{Z}] \mathbf{F}_i$$

with

$$F_{i,Jj} = (\mathcal{J} + \underline{\mathcal{Q}}_i) \mathbf{u}_i(\mathbf{x}_J) \cdot \mathbf{e}_j$$

4.4. Comments on the proposed methodology

By keeping all degrees of freedom (d.o.f.s) over the coupling domain Ω_δ rather than condensing equations on its boundary $\partial\Omega_\delta$, we avoid the calculation of constraint modes and condensed dynamic stiffness for the random heterogeneity as required by the usual sub-structuring method. The proposed formulation also explicitly isolates the random heterogeneity of the soil and its dynamic stiffness in such a way that it can be discretized independently from any consideration of the mean stratification of the background soil medium. The above Equation (20) can be inverted directly in conjunction with Monte-Carlo simulations of the operator $\delta\mathcal{Z}$, provided that the operators $\underline{\mathcal{U}}$ and $\underline{\mathcal{Q}}_i$ have been properly evaluated. Apart from the successive inversions for each trial, the main cost of the method lies in this calculation—performed only once for all Monte-Carlo simulations—by the classical sub-structuring technique. The size of the finite-element discretization of the random dynamic stiffness operator $\delta\mathcal{Z}$ is essentially driven by the statistical properties of the random perturbation fields $\delta\mathbf{D}^s(\mathbf{x})$ and $\delta\rho^s(\mathbf{x})$, and especially their (spatial) correlation length, which thus drives the size of the algebraic system of Equation (22) by the present discretization scheme. It remains usually quite reasonable for correlation length of the soil random perturbations of the same order as, or few divisions of, the foundation lengths, which is expected to be the most unfavorable case. Thus although matrices in Equation (22) are complex and full, this system can be repeatedly inverted without particular care. For all these reasons and in view of the complexity and scale of the contemplated problem, it is believed that the proposed procedure is as a whole much more efficient than classical sub-structuring methods, and all the more so full 3-D finite-element analysis of the complete system with or without the relevant absorbing boundaries.

We now turn our attention to a discretization of the random dimension of the dynamic stiffness operator $\delta\mathcal{Z}$ adapted to Monte-Carlo simulations.

5. REDUCTION OF THE RANDOM DYNAMIC STIFFNESS OPERATOR FOR PERTURBATIONS

The system equation (20) is now well suited to deal with a non-invertible dynamic stiffness operator, and more specifically a random one. This system is inverted along the random dimension by Monte-Carlo simulations, which require the construction of realizations of the random matrix $\delta\mathcal{L}$ from realizations of the stochastic fields $\delta\mathbf{D}^s(\mathbf{x})$ and $\delta\rho^s(\mathbf{x})$ modelling the random perturbations of the soil mechanical parameters in Ω_s . These perturbations being non-homogeneous stochastic fields defined on Ω_δ , the numerical cost of simulating their realizations can be prohibitive, so reduction techniques need to be implemented. This is done introducing a functional reduction of the covariance operators of the stochastic fields $\mathbf{P}^s$ and $\delta\rho^s$, based on their usual algebraic properties [27]. For that purpose, we also assume that $\int_{\Omega_\delta} R_\rho(\mathbf{x}, \mathbf{x}) d\mathbf{x} < +\infty$ and $\int_{\Omega_\delta} \text{Tr } \mathbf{R}_\mathbf{P}(\mathbf{x}, \mathbf{x}) d\mathbf{x} < +\infty$. These conditions mean that the realizations of the second-order stochastic fields $\mathbf{P}^s$ and $\delta\rho^s$ are also assumed square integrable on Ω_δ .

5.1. Karhunen–Loève expansion of the mechanical parameters perturbation fields

We form the covariance matrix function $\mathbf{R}_\delta$ (remind the fields are centred) of the random perturbations of the mechanical parameters in Ω_s given by

$$\mathbf{R}_\delta(\mathbf{x}, \mathbf{x}') = \begin{bmatrix} \mathbf{R}_\mathbf{P}(\mathbf{x}, \mathbf{x}') & \mathbf{R}_{\rho, \mathbf{P}}(\mathbf{x}, \mathbf{x}') \\ \bar{\mathbf{R}}_{\rho, \mathbf{P}}^T(\mathbf{x}, \mathbf{x}') & R_\rho(\mathbf{x}, \mathbf{x}') \end{bmatrix}$$

and defined on $\Omega_\delta \times \Omega_\delta$. It can be decomposed along its principal components $\{\boldsymbol{\Theta}_k\}_{k \geq 1}$ as [27]

$$\mathbf{R}_\delta(\mathbf{x}, \mathbf{x}') = \sum_{k=1}^{\infty} \theta_k \boldsymbol{\Theta}_k(\mathbf{x}) \otimes \bar{\boldsymbol{\Theta}}_k(\mathbf{x}') \quad (23)$$

where the eigenvalues $\theta_1 \geq \theta_2 \geq \dots \rightarrow 0$ are positive decreasing real numbers with finite multiplicity, possibly excepting 0, and the eigenvectors $\{\boldsymbol{\Theta}_k\}_{k \geq 1}$ form a complete orthonormal family. The Karhunen–Loève expansion of the perturbation random fields $\mathbf{P}^s$ and $\delta\rho^s$ is written

$$\begin{pmatrix} \mathbf{P}^s(\mathbf{x}) \\ \delta\rho^s(\mathbf{x}) \end{pmatrix} = \sum_{k=1}^{\infty} \xi_k \sqrt{\theta_k} \boldsymbol{\Theta}_k(\mathbf{x}) \quad (24)$$

where the $\{\xi_k\}_{k \geq 1}$ are complex-valued second-order random numbers, centred and uncorrelated

$$\mathbb{E}[\xi_k] = 0, \quad \mathbb{E}[\xi_k \bar{\xi}_{k'}] = \delta_{kk'} \quad (25)$$

Re-ordering the components of the eigenvectors $\{\boldsymbol{\Theta}_k\}_{k \geq 1}$ to extract the elasticity tensor coefficients (first components in the same order as for the definition of $\mathbf{P}^s$) and mass density (the last component), Karhunen–Loève expansions for the perturbation stochastic fields $\delta\mathbf{D}^s(\mathbf{x})$ and $\delta\rho^s(\mathbf{x})$ can be written as

$$\delta\mathbf{D}^s(\mathbf{x}) = \sum_{k=1}^{\infty} \xi_k \sqrt{\theta_k} \mathbf{D}_k(\mathbf{x}); \quad \delta\rho^s(\mathbf{x}) = \sum_{k=1}^{\infty} \xi_k \sqrt{\theta_k} \rho_k(\mathbf{x}) \quad (26)$$

where $\mathbf{D}_k$ and ρ_k , for $k \geq 1$, are deterministic elasticity tensors and densities, respectively, constructed from the components of the eigenvectors Θ_k . The series above are convergent in the mean square sense. Practically, they are truncated at an order K whose choice is driven by the relative error ε , in the mean square sense, we wish to achieve by these representations. The mean square error ε_K^2 up to the order K is

$$\varepsilon_K^2 = \sum_{k=K+1}^{\infty} \theta_k \leq \varepsilon \sum_{k=1}^{\infty} \theta_k$$

Thus K is chosen such that

$$1 - \frac{1}{\Pi} \sum_{k=1}^K \theta_k \leq \varepsilon \quad (27)$$

where $\Pi = \int_{\Omega_\delta} \text{Tr} \mathbf{R}_\delta(\mathbf{x}, \mathbf{x}) d\mathbf{x}$ is the total ‘power’ of the random field $(\mathbf{P}^s, \delta\rho^s)$.

5.2. Reduced formulation of the coupling for Monte-Carlo simulations

Using the Karhunen–Loève expansions (26) above up to an order K , we define dynamic stiffness operators $\{\mathcal{Z}_k(\omega)\}_{1 \leq k \leq K}$ replacing $\delta\mathbf{D}^s$ and $\delta\rho^s$ by $\sqrt{\theta_k}\mathbf{D}_k$ and $\sqrt{\theta_k}\rho_k$, respectively, in $\delta\mathcal{Z}(\omega)$, and such that

$$\delta\mathcal{Z}(\omega) = \sum_{k=1}^K \xi_k \mathcal{Z}_k(\omega) \quad (28)$$

Then we can rewrite the system Equation (20) in its reduced form along the random dimension

$$\left[\underline{\mathcal{U}} - \sum_{k=1}^K \xi_k \underline{\mathcal{U}} \mathcal{Z}_k \underline{\mathcal{U}} \right] \Phi = \sum_{k=1}^K \xi_k \underline{\mathcal{U}} \mathcal{Z}_k [\mathcal{J} + \underline{\mathcal{Q}}_i] \mathbf{u}_i \quad (29)$$

The randomness of the overall coupled problem is now supported only by random numbers, the weighting coefficients $\{\xi_k\}_{1 \leq k \leq K}$, which are much easier to simulate than the stochastic fields $\delta\mathbf{D}^s$ or $\delta\rho^s$. All matrices in the reduced system (29) are deterministic, so they only need to be computed once for all Monte-Carlo simulations. The numerical cost to pay for this simplification is the diagonalization of $\mathbf{R}_\delta$, which is performed by any standard algorithm [28].

5.3. Comments on the Monte-Carlo simulations

The reduced system (29) can also be inverted by introducing a Neumann expansion of the left-hand side matrix [18], or using a polynomial chaos projection [10]. These last two approaches are relevant only for Gaussian—or a known transformation [29] of Gaussian—stochastic fields, the random weighting coefficients being Gaussian as well in this case. However perturbations of mechanical parameters can hardly be modelled by Gaussian stochastic fields if the positiveness property has to be satisfied for the total elasticity tensor and mass density, since the support of the normal distribution is the whole real line. As we also want to consider possibly large perturbations, the Gaussian probability to have negative parameters is non negligible. One solution consists in checking, for each trial, the positiveness of the fields involved and reject those trials for which this property is not fulfilled. Recently, a methodology was proposed [30] to simulate non-Gaussian stochastic fields of a general class by Karhunen–Loève expansions, however it remains computationally rather costly. Here the random weighting

coefficients are chosen to follow a uniform distribution on $]-1, 1]$ with unit variance: the second-order properties of the perturbation random fields are well reproduced by this choice [27], but not their probability laws (which are in any case unknown *a priori*).

Note finally that uses of the reduction techniques by a Karhunen–Loève expansion and polynomial chaos projection are well established in the field of stochastic partial differential equations or signal analysis, but have received only recently a broad attention from the engineering mechanics community [10].

6. A VALIDATION CASE STUDY

In order to validate the sub-structuring method on volume interface developed above, we consider a very simple example in which it is compared with the classical sub-structuring technique on surface interface. The numerical comparisons described here are not intended, of course, to provide a full, rigorous proof of the mathematical problem considered. This would require a careful analysis of solvability and stability of the formulation in both the continuous and discretized cases, as well as the convergence of the numerical discretization scheme used. Such an analysis is out of the scope of this paper, and is not completely available yet.

The methodology has been implemented in the computer code `MISS-3D` [31] dedicated to multi-domains and solid–fluid interaction analysis by boundary integral equations and the boundary element method with analytical or numerical Green functions. This code has been extensively used for the analysis of large realistic case studies such as vibrations of arch dams and their water reservoirs under earthquake, seismic site effects for a complex topography, vibrations induced by moving loads, seismic borehole modelling, or waves in periodic heterogeneous media (see for instance References [26, 32]). These examples validate the classical sub-structuring technique on surface interface as implemented in `MISS-3D` so we can use it to construct the reference solution for our comparisons.

The geometrical layout is as follows (see Figure 3): an homogeneous inclusion occupying a cubic domain Ω_δ is embedded in an homogeneous half-space Ω_s just beneath a rigid massless square foundation loaded by a vertical constant point force $\mathbf{f}$ at its center. Its interface with the soil half-space is Γ . We compute the vertical displacements at three points A , B and C embedded in the half-space at the vertical of the force, normalized with respect to the static vertical displacement observed at A without inclusion. Contrasts between the mechanical parameters in the half-space and in the inclusion are characterized by the constant coefficients n_ρ and n_c such that

$$1 + \frac{\delta\rho^s}{\underline{\rho}^s} = n_\rho, \quad 1 + \frac{\delta c_p^s}{\underline{c}_p^s} = 1 + \frac{\delta c_t^s}{\underline{c}_t^s} = n_c$$

for, respectively, the compression and shear wave velocities $\underline{c}_p^s$ and $\underline{c}_t^s$ in the half-space without inclusion. The analysis here is thus restricted to deterministic perturbations. Since the inclusion is occupied by a deterministic homogeneous medium, the sub-structuring technique on surface interface described in Section 3 can be applied efficiently in this case, provided it is extended to multi-domains decomposition. Three sub-domains are involved here, the foundation, the inclusion and the embedding half-space; the interfaces are then $\Gamma \cup \partial\Omega_\delta$. The results obtained by this method are labelled ‘inclusion by BEM’ in Figure 4. Those obtained by the volume

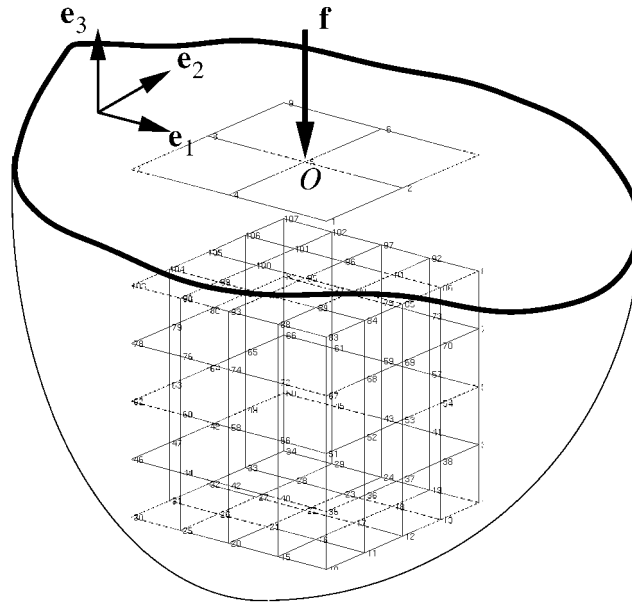


Figure 3. Validation case study: mesh layout for the inclusion discretized by 16 squares on each face.

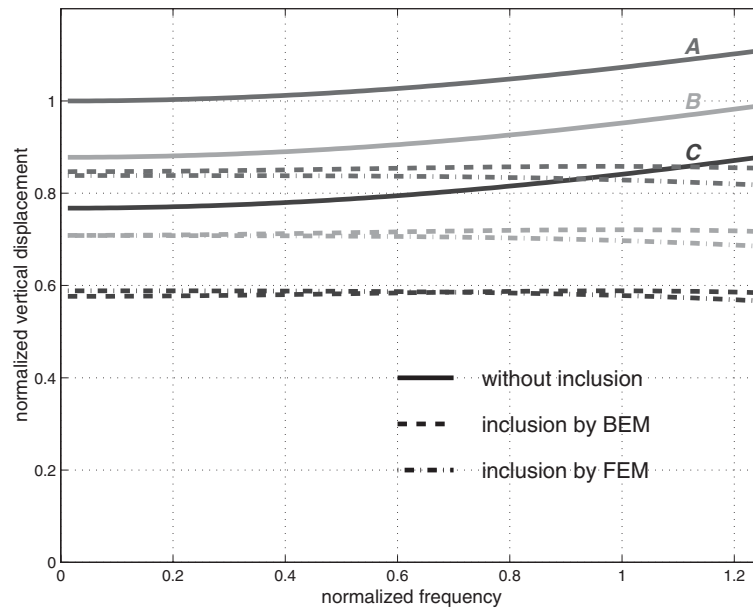


Figure 4. Validation case study: norm of the vertical displacements at points *A*, *B* and *C* for the inclusion discretized by 16 squares on each face.

interface approach are labelled ‘inclusion by FEM’ on the graphs, and we expect that similar results are given by the two methods. Of course, the sub-structuring technique on volume interface presents no particular advantage in this case, which is treated by such a method only for validation purposes. Its interest appears when the heterogeneity is random.

Figure 4 displays the normalized vertical displacements at A , B and C as functions of the normalized circular frequency $\omega e/c_t^s$ (e is the side length of the foundation) when the boundary $\partial\Omega_\delta$ of the inclusion is discretized by 16 square elements on each face (see geometry in Figure 3). Correspondingly, the inclusion volume is discretized by 64 hexaedres for the plots labelled ‘inclusion by FEM’. The calculations have been made for $n_c=2$ and $n_p=1$ (no perturbation of the mass). Displacements observed without inclusion have also been plotted as references. As can be seen from these graphs, the agreement between both methods is satisfactory. We also note that for this case, if N_e is the number of nodes per side of the mesh, the inclusion discretized ‘by BEM’ includes $6N_e^2$ nodes whereas it includes N_e^3 nodes when discretized ‘by FEM’. Thus, the number of d.o.f.s for both methods is comparable since $N_e=5$. Other extensive numerical tests, including the case of random heterogeneities, have been performed and may be found in Reference [19].

7. APPLICATION: SEISMIC SOIL–STRUCTURE INTERACTION IN UNCERTAIN ENVIRONMENT

The sub-structuring method on volume interface is now applied to a realistic example. For engineering purposes, we focus our analysis on the total displacement field in the structure, that is the diffracted field $\mathbf{u}_d$ in Ω_b since the free field is propagated only in Ω_s by the present formulation.

7.1. Geometry and layout

The structure we consider here is a nuclear power plant resting on a circular spread footing of radius $R \simeq 25$ m (see Figure 5 for the geometry and mesh layout). We restrict our analysis to the case of random Lamé’s moduli for the soil. This means that the elasticity tensor in the soil medium occupying Ω_s is isotropic and depends only on two parameters λ^s and μ^s written in the perturbation zone Ω_δ as

$$\lambda^s(\mathbf{x}) = \underline{\lambda}^s(z) + \delta\lambda^s(\mathbf{x}), \quad \mu^s(\mathbf{x}) = \underline{\mu}^s(z) + \delta\mu^s(\mathbf{x})$$

where $\underline{\lambda}^s$ and $\underline{\mu}^s$ are Lamé’s moduli for the soil mean stratification, function of the depth $z = \mathbf{x} \cdot \mathbf{e}_3$. The mass density of soils is known to vary smoothly with depth, so its perturbations are disregarded in this case. The deterministic free field considered is a vertically incident plane SH-wave. Examples of the analysis of the influence of a random free field for a similar structure can be found in Reference [19]. The structural displacements on a fixed basis are expanded along the first 30 eigenmodes, with constant critical damping rates $\zeta_A = 7$ per cent. The stratified soil medium (without heterogeneity) has three horizontal layers and the bedrock (layer #4 in Figure 5) whose characteristics are summarized in Table I. The corresponding Green tensors $\mathbf{U}^{Gs}$ and $\mathbf{T}^{Gs}$ of the stratified half-space are numerically computed by the reflexion/transmission coefficients method of Kennett [13]. The random soil

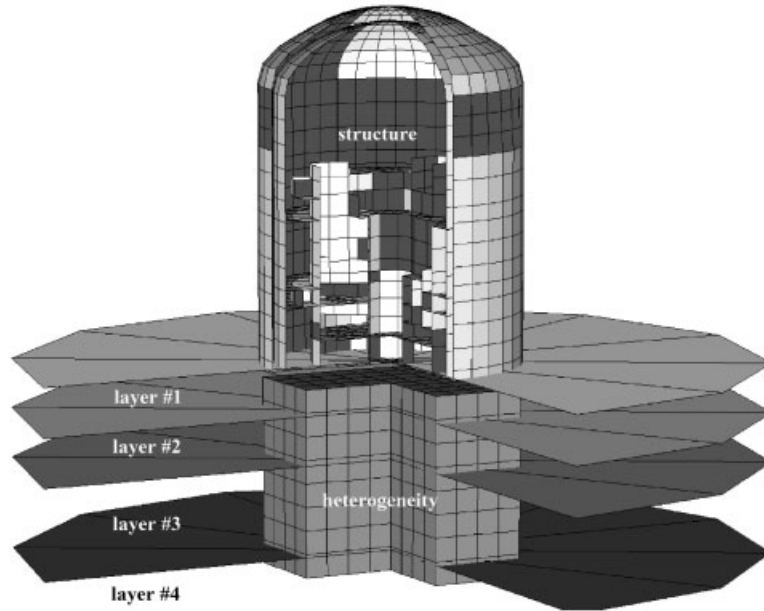


Figure 5. Example of a nuclear power plant on a stratified soil with random heterogeneity. As in Figure 3, only the soil–structure interface and the volume of the heterogeneity need to be meshed by the present methodology.

Table I. Characteristics of the stratified soil: thickness, Lamé's moduli and mass density (the hysteretic damping rate is 5 per cent).

	h (m)	$\underline{\lambda}^s$ (MPa)	$\underline{\mu}^s$ (MPa)	$\underline{\rho}^s$ (kg/m ³)
Layer #1	10.5	1.16×10^4	0.37×10^4	2400
Layer #2	10.5	1.95×10^4	0.76×10^4	2500
Layer #3	19.5	2.12×10^4	0.60×10^4	2400
Bedrock	∞	1.62×10^4	0.99×10^4	2500

heterogeneity occupies a cubic block $40 \times 40 \times 40$ m³ and its auto-correlation function is taken in the form

$$[\mathbf{R}_\delta(\mathbf{x}, \mathbf{x}')]_{pq} = \sigma_p \sigma_q \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|}{\ell}\right)$$

where $p, q \in \{1, 2\}$; σ_1 and σ_2 are the standard deviations of Lamé's moduli. As an illustration, we plot in Figure 6 the first four eigenmodes of such an exponential correlation operator for a real-valued random field ($p = q = 1$) defined on a square $]-e, e[\times]-e, e[$. The application is for $\sigma = 1$, $e = 5$ and $\ell = 2$.

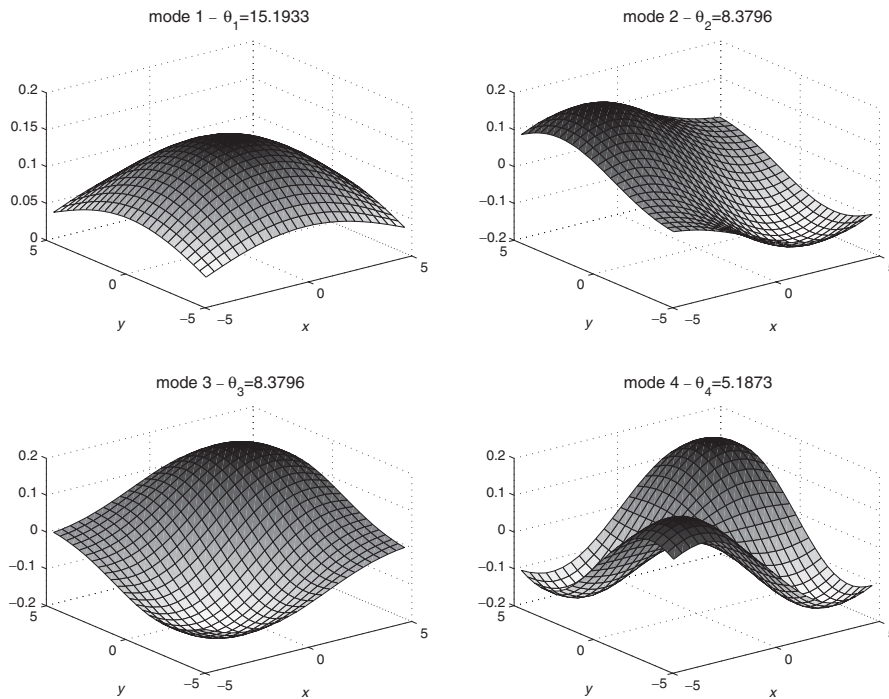


Figure 6. Karhunen–Loève eigenmodes $\Theta(x, y)$ for an exponential correlation operator on the square $]-e, e[\times]-e, e[$, with $e = 5$, $\ell = 2$ and $\sigma = 1$.

7.2. Results

Figure 7 displays the log-scaled amplitude of the horizontal displacement $u_x = \mathbf{u}^b \cdot \mathbf{e}_1$ on top of the structure normalized by the incident wave amplitude when: (i) at first, the correlation length ℓ for the soil is set to 50 m and the standard deviations $\sigma = \sigma_1 = \sigma_2$ of Lamé's moduli are of the order of 25 per cent; (ii) the correlation length is $\ell = 20$ m and the standard deviations of Lamé's moduli are of the order of 50 per cent. In the first case 500 trials and 5 modes were used in the Karhunen–Loève expansion, the relative error ε obtained being $\varepsilon = 11$ per cent (see Equation (27)); in the second case 1000 trials and 20 modes were used, the relative error being $\varepsilon = 8$ per cent. Mean values and confidence intervals are plotted, the latter being defined by the upper and lower envelopes such that $P(|u_x - \mathbb{E}[u_x]| < U_c) \geq p_c$ with a confidence level $p_c = 95$ per cent. The upper and lower envelopes are constructed using Chebychev's inequality [27] which results in $U_c(\omega)$ such that

$$p_c = 1 - \frac{V_{u_x}}{U_c^2}$$

where $V_{u_x}(\omega) = \mathbb{E}[|u_x(\omega) - \mathbb{E}[u_x(\omega)]|^2]$. The mean values for both cases are seen to be almost undistinguishable from the amplitude observed without inclusion; however the confidence regions are, as expected, quite different. From an engineering point of view, these

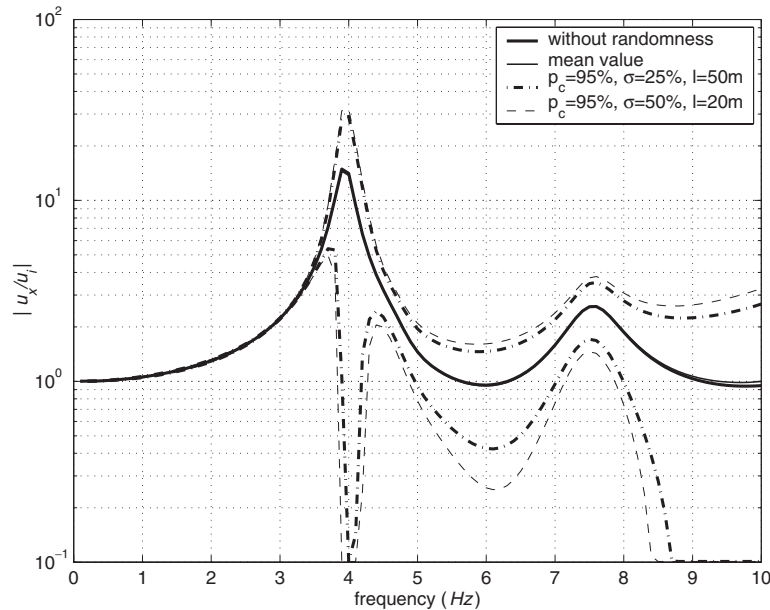


Figure 7. Normalized structural displacement under a vertically incident SH-wave for randomly perturbed soil stiffness, three stiff layers stratification: response without randomness (thick solid line), mean response (thin solid line), and confidence intervals for (i) $\ell = 50$ m and $\sigma = 25$ per cent (thick dashed lines), and for (ii) $\ell = 20$ m and $\sigma = 50$ per cent (thin dashed lines).

results show that a reduction of the structural vibration amplitude up to 5 per cent with $\sigma = 25$ per cent, and up to 10 per cent with $\sigma = 50$ per cent, can be obtained with a confidence of 95 per cent when the soil uncertainty underneath the foundation is accounted for. We also note that, for the present configuration, the fundamental eigenfrequency of the coupled soil–structure system is almost unchanged by the presence of random heterogeneities.

The soil described in Table I above is quite stiff as compared to usual soils. Other calculations have been performed with a softened soil having only one horizontal layer above the bedrock. The thickness of this layer is 40.5 m, its density is $\rho^s = 2400 \text{ kg/m}^3$, its Young's modulus is $E^s = 5 \times 10^3 \text{ Mpa}$ and its Poisson's coefficient is $\nu^s = 0.38$ (corresponding to $\lambda^s = 5.74 \times 10^3 \text{ MPa}$ and $\mu^s = 1.81 \times 10^3 \text{ MPa}$); the characteristics of the bedrock are unchanged from the previous example. Figure 8 is the same as Figure 7, except that now $\ell = 15$ m and $\sigma = 75$ per cent. 3000 trials were used for these parameters and $K = 40$ modes for the Karhunen–Loève expansion of the perturbations of Lamé's moduli (leading to $\varepsilon = 5$ per cent). In this example, soil–structure interaction effects are more pronounced than in the previous one thanks to the softening of the soil. The dispersion of the structural response for the same incident wave as in the previous case is increased, but qualitatively similar conclusions for the reduction factors can be drawn: a reduction of the structural vibration amplitude up to 15 per cent can be achieved with a confidence of 95 per cent. The fundamental eigenfrequency of the coupled system is almost unchanged as well, however slight discrepancies can already be observed for the higher mode at roughly 7.5 Hz. This trend agrees with the results obtained in the frame of the more general theory of symmetric positive-definite random matrices [11].

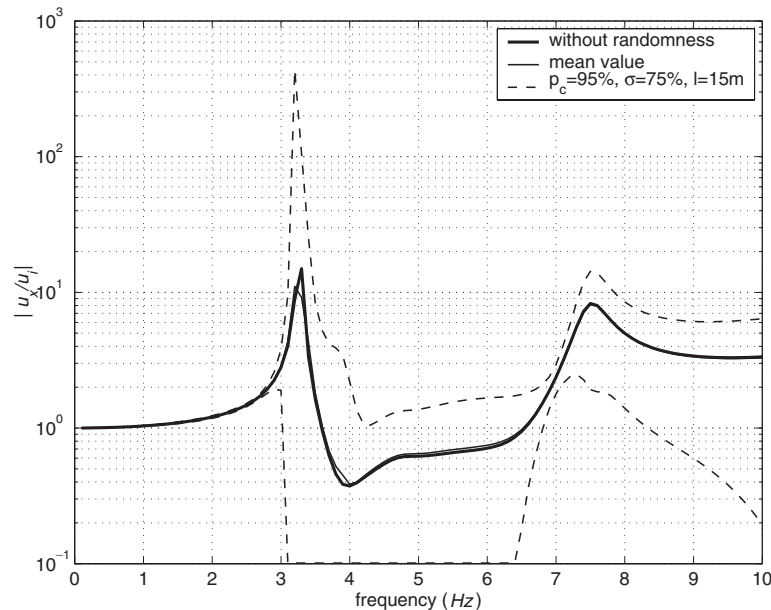


Figure 8. Normalized structural displacement under a vertically incident SH-wave for randomly perturbed soil stiffness, one soft layer stratification: response without randomness (thick solid line), mean response (thin solid line), and confidence interval for $\ell = 15$ m and $\sigma = 75$ per cent (thin dashed lines).

As regards reduction of structural vibration levels, comparable results, both qualitatively and quantitatively, were obtained by Toubalem *et al.* [33] by a simplified approach.

8. CONCLUSIONS

We have presented a continuum-based variational formulation of the coupling between a weakly damped structure and a stratified soil half-space exhibiting large random perturbations of its mechanical parameters over a bounded domain near the structural foundation. The coupling between the structure and the stratified soil medium without perturbation is written by a classical dynamic sub-structuring method, using boundary integral equations for the soil half-space. The coupling with the soil perturbations is written by an alternative sub-structuring procedure keeping all d.o.f.s over the perturbed domain. Its randomness is addressed in a much more efficient way using such a formulation, especially in view of the coupling with an unbounded domain. In addition, a functional reduction technique of the covariance operator of the perturbations has been used in order to discretize the random dimension of the coupled problem considered. The overall procedure described in this paper has been applied to a realistic complex case of soil–structure interaction analysis under earthquake for uncertain soil stiffness parameters. This very large 3-D model is treated efficiently with the numerical tools introduced, whereas more classical methods, such as the finite-element method, may lead to prohibitive computational costs and less accurate estimates. From an engineering point of view, this example shows that soil knowledge stochasticity is likely to induce lower

vibration amplitudes in large structures under earthquake. Dispersion of the structural response is increased for softer soils for which soil–structure interaction effects are more pronounced. It is also observed that the fundamental eigenfrequency of the coupled system is almost unchanged, but this may be no longer relevant for higher modes as the frequency is increased. These observations agree with other results published elsewhere in the literature.

Although presented here for two sub-domains, the proposed sub-structuring method on a so-called ‘volume’ interface can be readily extended to multi-domain analysis. It also allows for disconnected and/or overlapping FEM mesh for the heterogeneity, whereas a spectral discretization of the coupling force field can be used as done in Reference [17]. The present approach can be applied to account for random incident waves as well. Indeed, the structure response is linear with respect to the incident wave field through a scattering operator which can be shown to be an integral operator defined on the soil–structure interface. Therefore standard techniques can be implemented, for instance spectral analysis, or a functional reduction of the covariance operator of the incident wave on the soil–structure interface, similarly to the reduction introduced for the random perturbations of the soil mechanical parameters.

ACKNOWLEDGEMENTS

This study was carried out while the first author was affiliated to Laboratoire MSS-Mat, Ecole Centrale de Paris, Châtenay-Malabry. Financial support from Electricité de France EDF/DRD at that time is greatly acknowledged.

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